Project Proposal

DailyBasket Data Warehouse - Phase 1

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1. Introduction

DailyBasket is a grocery retail chain that sells food, drinks and household items to families and small shops. Sales come from its stores, a website and a mobile app. This project builds a complete Data Warehousing and Business Intelligence (DWBI) solution for DailyBasket, from raw data to dashboards that help managers decide.

2. Problem Statement

DailyBasket keeps its data in separate places: orders, customers, products, inventory and promotions. These systems are built to run daily work, not to answer big questions. Managers cannot easily see sales trends, best customers, slow products, weak regions or which promotions work. A data warehouse will bring all this data together in one clean, easy-to-query place.

3. Project Objectives

- Understand the business and write clear business requirements.
- Design a dimensional model (star schema) built around business processes and questions, not around copying operational tables.
- Build an ETL pipeline: ingest raw data, stage it, clean it and load the warehouse (Phase 2).
- Run OLAP analysis (roll-up, drill-down, slice, dice) with SQL (Phase 3).
- Build a Power BI dashboard that answers the business questions (Phase 3).

4. Business Processes

Business Process	What it is	Fact Table Type
Sales Transactions	Every product bought in an order	Transaction fact (fact_sales)
Inventory Snapshots	Stock level of each product at the end of each day	Periodic snapshot fact (fact_inventory_snapshot)
Promotion Events	Which product was part of which promotion on which day	Factless fact (fact_promotion_event)

5. Proposed Data Pipeline

Layer	What happens	Tool
Source Systems	Raw CSV files for orders, customers, products, inventory, promotions	CSV files / Python (Pandas) generated data
Data Ingestion	Load raw files as they are	Python / Pandas
Staging Layer	Raw copy kept in staging tables	PostgreSQL
Cleaning and Transformation	Fix nulls, duplicates, types; build keys	SQL / Pandas
Data Warehouse	Dimension and fact tables (star schema)	PostgreSQL
OLAP Analysis	Roll-up, drill-down, slice, dice queries	SQL

Layer	What happens	Tool
BI Reporting	Dashboards and reports	Power BI Desktop
Business Decisions	Managers act on the insights	-

6. Technology Stack

- **Database:** PostgreSQL with pgAdmin
- **Data work:** Python and Pandas
- **BI tool:** Power BI Desktop
- **Modeling diagrams:** Draw.io / dbdiagram.io
- **Version control:** GitHub (all SQL and scripts are kept so the project can be repeated)

7. Scope

In scope: sales, customers, products, regions, sales channels, inventory and promotions for DailyBasket (about 2 years of history).

Out of scope: real-time streaming, payment gateway data, supplier and HR data, machine learning.

8. Timeline and Marks

Phase	Weeks	Work	Marks
Phase 1 - Design	1 to 3	Business analysis and dimensional design	30%
Phase 2 - Implementation	4 to 6	Staging, ETL, warehouse build	40%
Phase 3 - Analytics and BI	7 to 8	OLAP, views and Power BI dashboard	30%

9. Phase 1 Deliverables

- Project Proposal (this document)
- Business Requirements Document
- 10+ Business Questions
- Fact and Dimension Identification
- Grain Definition Document
- Star Schema Diagram (PNG/PDF)
- Data Dictionary (Excel)

10. Expected Outcome

A portfolio-quality DWBI project that follows the full journey: data warehouse foundations, dimensional modeling, ETL, OLAP and BI reporting, using only free tools.